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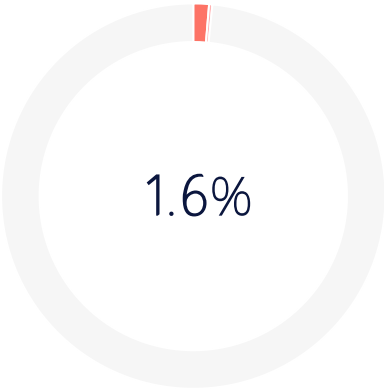

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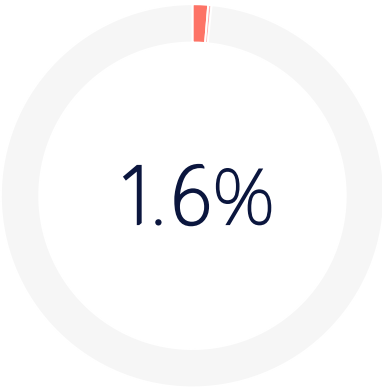

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1. IntroductionVirtual reality (VR) has become an incredibly effective medium for producing immersive andinteractive environments that create a strong feeling of being present there and transcend ordinarydigital interfaces by enveloping users with computer-created worlds[Sla18, FRR+17]. Thisunique capability has led to its adoption not only in entertainment, but also in education,training, therapy, and healthcare applications [Bak19]. Virtual reality has been increasinglyseen as a tool for strengthening mental health and wellness because it provides protected anddistraction-free spaces where users can retreat from external stressors and engage in relaxationpractices [R+21, Bai20].In conjunction with such advances in technology, meditation has traditionally been used as atool to develop relaxation, improve emotional regulation, and reduce anxiety and stress[SES+12,LSDD08, Man11]. Scientific evidence illustrates the ability of meditation to help resilience andpromote general mental health. Extensive research shows that meditation can reduce stress andanxiety, improve concentration, and improve overall well-being [G+14, Z+15]. However, manypeople struggle to maintain a regular meditation regimen due to distractions, lack of guidance,difficulty staying focused, or inability to relax the mind.VR offers a very useful bridge; it can turn proven relaxation and mindfulness methods intoengaging sensory experiences that make it easier for people to start and keep practicing. Thecombination of the two spaces, meditation in VR spaces, has created VR-centered meditationapplications. These systems present serene and immersive environments that can facilitate theuser's ability to transcend peripheral interruptions and focus more intensely on their meditation[LGM18, WWM20]. However, most commercial VR meditation solutions are unchanging,presenting pre-programmed content that is not dynamically adjusted based on the user's state ofemotion or physiology. This static responsiveness can inhibit their ability to facilitate increasedrelaxation and prolonged focus. In order to fill the gap, this thesis is motivated by the opportunityto design and evaluate an interactive, AI-powered, voice-mediated VR meditation system whichcan guide through meditation based on user stress level.1.1. Topic DescriptionThis thesis investigates a voice-interactive, AI-guided VR meditation system developed inUnity 3D for HTC VIVE Pro. The application delivers natural-language voice commands(generated by a GPT) to guide breathing, attention control, and relaxation strategies, whileadaptive audiovisual transitions (e.g. brightening and fading of light during eyes-closed phasesto help with breathing, along with meditation music) shape the meditative ambience.To ground personalisation, the user will be asked about their stress level and the guidance willbe given according to that level. The system also records physiological signals (SCL) Skin4

1. IntroductionConductance Level, (SCR) Skin Conductance Response, Skin Temperature, (BVP) BloodVolume Pulse, (PVA) Pulse Volume Amplitude, Pulse / Heart Rate, Motion / Motility andHeart Rate variability (HRV) derived from BVP–using Biofeedback 2000 Xpert sensors.These data is collected during the session and later is used for the comparison of the outcomeof meditation[LG13a, CS18]. A user study with >40 participants compares the AI-guidedVR meditation against a conventional (non-VR) meditation baseline, combining physiologicaloutcomes with self-reports of relaxation and affect.1.2. Research Questions and HypothesesRQ1. Does AI-guided voice interaction combined with adaptive audiovisual cues improverelaxation and mindfulness during VR meditation compared to a conventional baseline?H1. Participants testing the AI-guided VR meditation will exhibit significantly greater post-session relaxation (physiological and self-report) than conventional meditation.RQ2. Do "eyes-closed mode" visuals (e.g., breathing light effect presented during instructedeyes-closed phases) enhance the perceived depth of meditation and calmness?H2. Participants will report higher immersion and calmness during eyes-closed phases aug-mented by soothing visuals and ambient audio.RQ3. Can a combination of physiological (HRV, skin conductance etc) and self-report measuresprovide valid, convergent indicators of meditation effectiveness in an AI-guided VR environment?H3. Physiological improvements (e.g., increased HRV, reduced electrodermal activity) willcorrelate with higher self-reported relaxation and positive affect, indicating convergent validity.1.3. Scientific Contributions- Design Contribution: A multimodal, AI-guided voice and audiovisual-adaptive VRmeditation prototype that personalizes ambience and guidance to support relaxation,building on best practices in user-centered VR design.- Methodological Contribution: An evaluation protocol combining physiological (HRV,skin

conductance) and psychological (self-report) measures to assess meditative outcomesin VR.- Empirical Contribution: Evidence on whether AI-guided voice prompts and adaptiveaudiovisual transitions improve relaxation relative to a conventional meditation baseline,including analysis of “eyes-closed mode” visuals.- Synthesis Contribution: A literature-based rationale connecting meditation science,biofeedback, and adaptive interaction in VR, clarifying where adaptive intelligence addsvalue beyond static experiences.2007-12-31/100/IN00/3333 5

1. Introduction1.4. Chapter OutlineThe remainder of this thesis is organised as follows. Chapter 2 reviews related work on virtualreality meditation, biofeedback-based stress regulation, AI-driven adaptivity, and multimodalinteraction design, and identifies the research gap that motivates the present study. Chapter 3 de-scribes the methodology, including the experimental design, hardware and software architecture,system implementation, participant procedures, measurement instruments, and data processingpipeline. Chapter 4 presents the empirical results, covering both physiological responses andquestionnaire outcomes, and compares the effectiveness of conventional versus AI-guided VRmeditation. Chapter 5 summarises the main findings, answers the research questions, andsituates the results within the broader literature. Chapter 6 outlines limitations and proposesfuture research directions for adaptive, biofeedback-aware VR meditation systems.2007-12-31/100/IN00/3333 6

2. Literature Review2.1. IntroductionThe purpose of this literature review is to provide an overview of existing research on the topicsof virtual reality, meditation, artificial intelligence, and physiological tools, such as biofeedback.This synthesis highlights theory and findings that form the foundation for understanding howVR can enhance mindfulness practices through emerging technology like artificial intelligence.The review is organized thematically to guide the reader from fundamental concepts such asevolution and applications of VR to areas more specific to the field of meditation and to theregions where Biofeedback mechanisms and AI-driven systems are used. These studies revealhow integrating technological innovation with meditative practice can improve mental andemotional well-being.2.2. Virtual Reality: Concepts and ApplicationsVirtual Reality (VR) is a computer-generated environment that immerses users in realistic orimagined scenes and permits interaction through multisensory input and output [Jer16]. Twoconcepts are central to VR: immersion, the extent to which the sensory system is engaged by thevirtual environment, and presence, the subjective feeling of “being there” in the virtual world[Sla18, Bry96]. High levels of presence and immersion can significantly influence cognition,behavior, and emotion.While VR initially gained prominence in gaming and entertainment, it is now widely applied inhealthcare, psychotherapy, assessment, training, and education [FRR+17, R+21, Bak19]. Meta-analytic and review work on VR in mental health indicates that VR exposure and therapeuticenvironments can be effective for anxiety disorders, psychosis, and trauma-related conditionswhen integrated into structured treatment protocols [FRR+17, R+21]. Human-centered designguidelines emphasize usability, comfort, and emotional safety as key requirements for VRapplications targeting vulnerable or clinical populations [Jer16].More recent studies have explored the integration of VR with AI and adaptive interfaces ineducation and training contexts, for example VR classrooms with AI-generated scenarios orintelligent tutoring and assessment [KSKS24, Liu24, EYKC25]. In medical and proceduraltraining, VR has also been used to simulate complex interventions and measure stress responsesand user perceptions during high-stakes scenarios [RLGCZR+25]. Together, these findingsposition VR as a versatile medium for both skill development and emotional or stress-relatedinterventions.7

2. Literature Review2.3. Meditation and Psychological Well-BeingMeditation has a long-established role in promoting psychological balance and emotional re-silience. It encompasses diverse practices–such as mindfulness, recitation, and focused atten-tion–each offering unique physiological and cognitive benefits [SES+12, LSDD08, Man11].Meta-analytic evidence shows that meditation programs produce small to moderate improve-ments in psychological stress, anxiety, depression, and overall well-being across diverse popu-lations [G+14, SES+12].Theoretical and neuroscientific studies further reveal that meditation can strengthen attentionalregulation, enhance metacognitive monitoring, and modulate emotion-related brain circuits,particularly in the prefrontal cortex, anterior cingulate cortex, insula, and amygdala [LSDD08].Classic conceptual models such as de-automatization describe how meditation disrupts habitualcognitive and perceptual patterns, enabling more flexible and less reactive responses to stressors[Dei66, Man11].Despite its benefits, many individuals find it difficult to initiate and maintain a regular meditationpractice due to time constraints, lack of guidance, or difficulties in sustaining

attention. Re-cent studies have therefore investigated technology-assisted meditation, including smartphoneapps and VR-based systems, to provide more engaging, structured, and context-aware support[TVC18, CBN21]. These approaches aim to bridge the gap between traditional contemplativepractices and contemporary digital lifestyles.2.4. Virtual Reality Meditation: Integrating Technology andMindfulnessThe combination of VR and meditation, often called VR meditation, represents a novel approachto deepening relaxation and focus through interactive, sensory-rich environments [LGM18].VR meditation combines immersive virtual environments with guided meditative or relaxationtechniques. By presenting calming natural scenes or abstract visualisations along with audioguidance, these systems aim to deepen focus, reduce external distractions, and enhance the sub-jective quality of the meditative state [LGM18, TVC18]. Empirical work shows that immersiveVR nature environments can foster relaxation, reduce negative affect, and increase subjectivewell-being compared to control conditions [LGM18].Specific VR meditation protocols have been evaluated for anxiety reduction and mood im-provement, where participants undergo sessions of guided breathing and attention trainingembedded in virtual nature or restorative environments [TVC18, BRB+19]. Neuroadap-tive and biofeedback-enhanced VR systems further refine this approach by adjusting audio-visual content in real time according to physiological signals such as heart rate or respiration[KSJ+16, WWM20, XGH25]. Movement-based systems like LOOP combine body movementwith breath and attention to support engagement [FYM20], while mantra-based VR meditationhas been explored for stress reduction and EEG/HRV modulation [GKG+24].However, recent clinical and applied studies extend VR meditation to specific populations.For example, VR-based mind-body interventions have been tested for adolescent insomnia,chronic pain in older adults, and nursing students' stress and sleep quality [ZYK+22, SEB22,2007-12-31/100/IN00/3333 8

2. Literature ReviewKHK24]. Virtual reality meditation "retreats" and structured multi-session programs have alsobeen proposed as scalable tools to deliver contemplative practices to people who might not accesstraditional retreats [MZXY22]. However, many current applications still rely on pre-scriptedcontent and do not adapt to moment-to-moment physiological or emotional changes.2.5. Physiological Biofeedback and RegulationPhysiological biofeedback plays a central role in understanding and measuring relaxation. Phys-iological biofeedback involves measuring signals such as heart rate variability (HRV), pulse,skin conductance, and temperature and presenting aspects of these signals back to the user inreal time. The goal is to enhance awareness of internal states and support self-regulation ofstress and arousal. HRV biofeedback in particular trains individuals to breathe at a resonancefrequency that optimizes baroreflex functioning and promotes parasympathetic activity, leadingto calmer states and improved emotion regulation [LG13b].Electrodermal activity, typically captured as skin conductance level (SCL) and skin conductanceresponses (SCR), reflects sympathetic nervous system activity and is a sensitive indicator ofemotional arousal [Bou12]. In human-computer interaction and physiological computing,combinations of HRV, EDA, respiration, and peripheral temperature are used to infer mentalworkload, engagement, and stress [Fai09]. Integrating these measures into interactive systemsallows designers to create closed-loop experiences that respond to users' internal states.VR-based biofeedback systems combine immersive visual or auditory environments with real-time physiological feedback. Studies suggest that VR biofeedback can enhance relaxation,presence, and self-efficacy compared to non-VR or non-biofeedback conditions [WWM20,XGH25]. For instance, virtual environments may change colour, brightness, or motion based onHRV or breathing patterns, reinforcing relaxation through intuitive metaphors [CS18, BJN20].Systematic reviews underline the importance of standardized protocols and signal processingpipelines to ensure reliable findings across biofeedback studies [App20].Recent work has also explored the use of biofeedback in emotionally intense or stressful vir-tual experiences, such as fear-evoking scenarios or high-stakes medical simulations [HCGR18,RLGCZR+25, Rea21]. These studies illustrate how the same physiological channels used tomeasure stress can be leveraged to deliver adaptive feedback and support self-regulation training.2.5.1. Stress and Frustration Detection through Physiological SignalsStress and frustration arise when cognitive, emotional, or social demands are perceived as ex-ceeding coping resources. While self-report tools such as the PSS-10 or NASA-TLX capturesubjective evaluations of stress and workload, physiological signals provide continuous, objec-tive markers of autonomic nervous system activation. Under acute stress, sympathetic activationtypically leads to increased heart rate, reduced HRV, elevated skin conductance, and reducedperipheral temperature due to vasoconstriction [HP05, MS01, Bou12].Heart rate variability reflects the dynamic

balance between sympathetic and parasympatheticactivity; lower HRV is often associated with heightened stress and reduced self-regulatorycapacity [MS01]. Electrodermal activity is widely used as a direct index of sympathetic arousal,

(MAAS) measures dispositional mindfulness, defined as open orreceptive awareness of and attention to present-moment experience [BR03].Emotional valence is assessed using the Positive and Negative Affect Schedule (PANAS),which includes two independent subscales for positive and negative affect [WCT88]. Statemindfulness during or immediately after the meditation session is measured with the StateMindfulness Scale (SMS), which captures awareness of mind and body [TB13].In the VR condition, presence is evaluated with the I-Group Presence Questionnaire (IPQ),which measures spatial presence, involvement, and experienced realism [SFR01]. Cybersick-ness symptoms are assessed using the CSQ-VR, a validated questionnaire covering nausea,disorientation, and oculomotor discomfort in VR environments [KCDM19].Perceived workload is measured with the NASA Task Load Index (NASA-TLX), providingratings along six dimensions: mental, physical, and temporal demand, performance, effort,and frustration [HS88]. Overall usability and experiential quality are evaluated with the UserExperience Questionnaire–Short (UEQ-S), which yields pragmatic and hedonic qualityscores [SHT17]. Together, these instruments provide a multidimensional assessment of stressreduction, mindfulness induction, emotional state, cognitive workload, and experiential quality.2007-12-31/100/IN00/3333 11

2. Literature Review2.9. Commercial Applications and Research GapCommercial VR meditation platforms such as TRIPP, Guided Meditation VR, and FlowVRhave made immersive mindfulness tools more publicly accessible and illustrate the growinginterest in VR for emotional self-care and stress management [GZB+25]. Pilot trials with youthand other vulnerable groups show that VR meditation can be feasible and acceptable, and mayreduce anxiety or distress in challenging life situations [CKS+20]. Clinical and applied studiesfurther indicate that VR-based meditative interventions can help modulate arousal and improvesleep or pain-related outcomes [ZYK+22, SEB22, KHK24].However, reviews of commercial and research prototypes alike note that most VR meditationsystems still rely on static scenarios and linear, pre-scripted guidance [GZB+25]. They rarelyintegrate real-time physiological monitoring or AI-driven personalisation, and they often lackmechanisms for adapting content to users' momentary stress level, workload, or frustration.Recent work on VR stress measurement and training in medical and educational contextsemphasizes the importance of combining psychophysiological measures with user-centereddesign to truly understand and support users under stress [RLGCZR+25, Rea21].This literature therefore identifies a clear research gap: the need for AI-guided, biofeedback-informed VR meditation systems that can adapt instructions and audiovisual parameters dy-namically based on physiological and self-report indicators. Such systems have the potential todeliver more individualized, evidence-based relaxation experiences and to advance the clinicalreliability and scalability of digital mindfulness interventions.In summary, VR provides immersive, controllable environments capable of supporting mentalhealth interventions, including meditation and stress management. Meditation itself has arobust evidence base for improving psychological well-being, and VR-based meditation systemsextend these practices into interactive, engaging formats. Physiological biofeedback, particularlythrough HRV and electrodermal activity, offers objective insight into stress and relaxation, whileAI and multimodal design enable adaptive, personalized VR experiences.Despite promising developments, current VR meditation applications rarely combine AI-drivenguidance with real-time physiological adaptation. Addressing this gap, the present work focuseson the design and evaluation of an AI-guided, biofeedback-based VR meditation system thataims to reduce stress and enhance emotional regulation by responding dynamically to users'physiological and self-reported states.2007-12-31/100/IN00/3333 12

3. Methodology3.1. IntroductionThis chapter describes the methodological framework used to examine the effects of an AI-guided virtual-reality (VR) meditation system on users' stress, mindfulness, affect, workload, andoverall experience. It outlines the research design, participant characteristics, hardware-softwaresetup, experimental procedure, measurement instruments, data collection and processing steps,statistical analyses, and ethical considerations.3.2. Research DesignThe primary aim of this study was to compare the effectiveness of two meditation modalities inreducing stress after a common stress induction: (a) conventional audio-based meditation and(b) AI-guided virtual reality (VR) meditation.A mixed experimental design was adopted, combining:- a between-subjects factor: meditation modality (conventional vs. AI-guided VR), and- a within-subjects factor: experimental phase (stress induction, meditation).All participants first completed a resting baseline phase, followed by a stress-induction phaseusing the game Flappy Bird, and then a meditation phase.The

Flappy Bird task was used to elevate arousal and induce a comparable stress level acrossparticipants before the relaxation intervention. Participants were then randomly assigned toeither the conventional meditation condition or the AI-guided VR meditation condition, wherethe actual stress reduction was expected to occur.Physiological signals (heart rate, heart-rate variability, skin conductance, skin temperature, pulsevolume amplitude) were recorded continuously across all phases using the Biofeedback 2000Xpert system. Stress reduction was operationalized as the change in physiological indices fromthe stress phase to the meditation phase (e.g., decrease in SCL and heart rate, increase in HRV).These changes were compared between the two meditation groups to determine which modalityproduced stronger recovery from the induced stress.Self-report questionnaires were collected only after the meditation phase; therefore, subjectiveoutcomes follow a between-subjects comparison only, while physiological responses supportboth within-subject (phase-based) and between-subject (modality-based) analyses.13

3. Methodology3.3. Experimental Setup3.3.1. Rationale for Using Virtual Reality Instead of 2D DisplaysA central design decision in this study was the use of a fully immersive Virtual Reality (VR)environment rather than a conventional 2D display such as a television, monitor, or mobile device.This choice is grounded in both theoretical and empirical evidence on how immersion, attentionalfocus, and sensory isolation influence the quality of meditation and emotional regulation.VR offers the advantage of high presence, defined as the subjective feeling of “being there,”which is strongly associated with deeper engagement and reduced susceptibility to externaldistractions [Sla18, Jer16]. In contrast, 2D displays allow continuous exposure to the surroundingenvironment, ambient noise, and visual disturbances, which can interfere with meditative focus.Furthermore, immersive VR environments provide full-field visual occupation and spatial au-dio, creating a controlled sensory envelope ideal for guiding relaxation and regulating breathing.Prior research demonstrates that VR meditation enhances emotion regulation, reduces anxiety,and improves attentional stability more effectively than non-immersive media [TVC18, LGM18].For the AI-guided component, VR also enables embodied interaction–the user perceives AI-guided voice prompts as originating from within the virtual space, creating stronger perceivedconnectedness and therapeutic engagement compared to listening through a TV or phone speaker.Given these benefits, VR was selected to maximize immersion, minimize distraction, and createan optimally controlled environment for studying stress reduction and physiological regulation.Table 3.1.: Comparison between VR-based meditation and 2D (TV/Laptop) meditation.Feature VR Meditation (HTC VivePro)2D Meditation (TV/Laptop)Immersion High immersion via stereo-scopic 3D, spatial audio, andpresenceLow immersion; flat screenwith no spatial congruenceDistraction Level Low: blocks external visualstimuliHigh: environment remainsvisibleEnvironmental Control Complete simulated environ-mentEnvironment cannot be con-trolledEmotional Engagement Stronger due to embodiedpresenceModerate/weak emotional en-gagementAttentional Focus Deep attentional capture, iso-lation from surroundingsAttention is frequently dis-rupted by external cuesSuitability for AdaptiveScenesHighly effective, real-timeadaptation visible and impact-fulLimited, adaptation less no-ticeable on 2D displaysMeditation Depth Higher due to reduced mind-wanderingLower, more interruptionsFurthermore, VR environments support adaptive scene transitions (e.g., breathing visuals) thatalign dynamically with AI-guided instructions. Such adaptive cues are less perceptually impact-2007-12-31/100/IN00/3333 14

3. Methodologyful when delivered on a standard screen. For these reasons, VR is the most suitable medium fordelivering a controlled, immersive, and emotionally effective meditation experience.3.3.2. Hardware materialsThe immersive experience was delivered using an HTC Vive Pro headset. Physiological signalswere captured via Biofeedback 2000 Xpert sensors. A high-performance workstation ensuredstable rendering and real-time data processing.HTC Vive Pro Head-Mounted DisplayThe virtual reality condition was delivered using the HTC Vive Pro head-mounted display(HMD), a professional-grade VR system offering high-resolution stereoscopic visuals and pre-cise motion tracking. The headset features dual AMOLED displays with a combined resolutionof 2880 × 1600 pixels (1440 × 1600 per eye), a 90 Hz refresh rate, and a field of view ofapproximately 110° [HTC24]. These specifications help provide a smooth and immersive visualexperience with minimal motion blur, which is critical for sustaining presence during meditationtasks.The Vive Pro uses room-scale Lighthouse tracking, enabling accurate positional and rotationalmovement detection with sub-millimeter precision. Integrated stereo headphones provide spatialaudio, supporting auditory immersion for AI-generated meditation prompts. The ergonomicweight distribution

and foam padding contribute to prolonged comfort, reducing distractionsduring longer meditation sessions.Although the device supports integrated eye-tracking modules in extended variants, the presentstudy used the standard Vive Pro configuration without eye tracking. The headset was operatedvia SteamVR and connected to a Unity-based VR application developed for the experiment.This setup ensured consistent environmental rendering, stable frame timing, and reduced motionartifacts, all of which are important for meditation-based VR interventions.Figure 3.1.: HTC VIVE Pro head-mounted display.Biofeedback 2000 Xpert SystemPhysiological data were recorded using the Biofeedback 2000 Xpert system (Schuhfried GmbH)[Sch24], a professional psychophysiological measurement platform widely used in clinical,2007-12-31/100/IN00/3333 15

3. Methodologysports, and cognitive research. The system supports high-resolution acquisition of autonomicnervous system signals including cardiovascular, electrodermal, and thermoregulatory activity.It provides robust sensor modules, real-time feedback capabilities, and precise synchronization,making it suitable for meditation, stress-induction, and biofeedback interventions.Figure 3.2.: Biofeedback 2000 Xpert system used for physiological recording.The device consists of a wireless base unit communicating with individual sensor modulesthrough proprietary low-latency channels. Sampling rates vary between 10-200 Hz dependingon sensor type, allowing the system to capture both slow-varying signals (e.g., temperature)and fast dynamic responses (e.g., pulse waveforms or skin conductance responses). The systemsoftware provides real-time visualization and logging, and all data streams are automaticallytime-stamped to facilitate post-hoc synchronization with experimental events.The following physiological signals were recorded:Blood Volume Pulse (BVP). The BVP sensor uses photoplethysmography to detect volumetricchanges in peripheral blood flow. The waveform provides:- instantaneous heart rate (HR),- inter-beat interval (IBI),- heart-rate variability (HRV) indices (e.g., RMSSD),- pulse amplitude and vascular tone.BVP is highly sensitive to parasympathetic activity and is widely used in meditation and relax-ation research.Skin Conductance Level (SCL) and Skin Conductance Response (SCR). Electrodermalactivity (EDA) was measured via two electrodes attached to the fingers. The tonic component(SCL) reflects baseline sympathetic arousal, whereas phasic SCR peaks reflect moment-to-moment emotional or cognitive activation [Bou12]. SCL and SCR are among the most reliableindices of stress and frustration.2007-12-31/100/IN00/3333 16

3. MethodologySkin Temperature (TEMP). Temperature sensors measure peripheral vasodilation and vaso-constriction. Under stress, sympathetic activation reduces skin temperature, whereas relaxationincreases it. TEMP changes complement BVP and SCL in assessing autonomic balance.Pulse Volume Amplitude (PVA). PVA represents the strength of the pulse waveform capturedthrough the photoplethysmograph. It indicates peripheral blood flow and vessel dilation. LowerPVA signals may indicate stress-related vasoconstriction, whereas higher amplitudes suggestrelaxation.Pulse Frequency. Pulse frequency corresponds to the number of pulse peaks per minute. It isderived from the BVP waveform and provides a stable estimate of heart rate during meditation,stress induction, and recovery phases.Technical Operation The Biofeedback 2000 Xpert uses modular sensing units that transmitphysiological signals to a central receiver in real time. Each module contains:- analog-to-digital conversion,- noise filtering (including notch and band-pass),- signal amplification,- wireless data transmission.Data streams are processed through the system's software suite, which handles:1. real-time signal visualization,2. timestamping of events,3. high-frequency sampling,4. artifact detection,5. logging to CSV for offline analysis.The system maintains precise synchronization with external events through:- internal system clock,- session start markers,- trigger annotations inserted during VR scene transitions and meditation phases.This ensures that meditation onset, stress induction, AI prompt onset, and physiological signalscan be aligned accurately during statistical analysis.Physiological Parameters Analyzed The following features were extracted according to es-tablished psychophysiological guidelines:- Heart Rate (HR) and Inter-beat Interval (IBI) from BVP.- HRV (RMSSD) as a parasympathetic relaxation indicator.- Skin Conductance Level (SCL) as a tonic sympathetic arousal index.2007-12-31/100/IN00/3333 17

3. Methodology- Skin Conductance Responses (SCR peaks) during stress induction.- Pulse Volume Amplitude (PVA) to assess vascular relaxation.- Skin Temperature (TEMP) as a measure of vasomotor recovery.These metrics provide converging evidence of stress, cognitive load, frustration, and relaxationduring the experiment. HRV reflects parasympathetic activation [MS01], SCL reflects sym-pathetic arousal, and temperature and PVA reflect peripheral vascular changes. Together, theyoffer a comprehensive view of autonomic nervous system

regulation.Workstation All VR sessions and data processing were performed on a high-performanceworkstation with the following specifications:Component SpecificationCPU Intel Core i9-11900K (3.50 GHz)RAM 128 GB DDR4GPU NVIDIA GeForce RTX 3090 Ti (24 GB)Storage 16.4 TB (SSD + HDD)System Windows 10, 64-bitTable 3.2.: Workstation configuration used for VR rendering and data analysis.3.3.3. Software and EnvironmentThe implementation of the VR meditation system and the overall experimental workflow requiredseveral software components working together across content creation, artificial intelligenceprocessing, physiological signal acquisition, and data analysis. The following subsectionsdescribe the main software tools and frameworks used in this study.Unity 3D EngineThe virtual environment and meditation application were developed using the Unity 3D gameengine (Version 2021.x). Unity was selected due to its stability, extensive VR support, andability to integrate custom interaction logic, AI-driven components, and external hardware.The application includes the VR scene, audio pipeline, visual transitions (light fading, skyboxchanges), and user interaction logic. Unity's coroutine system and animation curves were usedto implement smooth temporal effects such as lighting transitions and ambient visual changes.SteamVR / OpenXR RuntimeVR rendering and headset tracking were enabled using the SteamVR framework with theOpenXR backend, which ensured compatibility with the HTC Vive Pro headset. SteamVRmanages:- stereoscopic rendering,

- Lighthouse positional tracking,- headset orientation updates,- frame-timing and reprojection.This ensures low-latency head tracking and stable frame rates, both critical for immersion andpreventing cybersickness during meditation.OpenAI GPT and Text-to-Speech IntegrationMeditation guidance in the VR condition was generated in real time using the OpenAI GPTlanguage model, accessed via the official Chat Completions API [Ope25]. During the session, theUnity application sent short cue prompts (e.g., breathing instructions and supportive phrases)together with the participant's reported stress level to GPT and received concise, calmingguidance sentences in return.These sentences were immediately forwarded to an online Google Text-to-Speech (TTS) service,which converted the text into audio. The resulting audio streams were played back inside the VRscene through Unity's AudioSource component, providing a continuous sequence of adaptive,AI-generated meditation prompts. A more detailed description of this pipeline is provided inSection 3.4.1.Biofeedback 2000 Xpert SoftwarePhysiological signals were recorded using the Biofeedback 2000 Xpert software suite, whichsupports real-time acquisition of:- Blood Volume Pulse (BVP),- Skin Conductance Level (SCL),- Skin Temperature (TEMP),- Pulse Volume Amplitude (PVA),- Pulse Frequency.The software automatically time-stamps all signals and logs them to CSV for offline analysis.Session markers corresponding to the experimental phases (baseline, stress induction, medita-tion) were inserted manually during data collection.Python for Data ProcessingPost-experiment physiological data processing was conducted in Python using the followinglibraries:- pandas – preprocessing and time alignment,- numpy – numerical operations,- neurokit2 – HRV extraction (RMSSD, IBI, peak detection),

- matplotlib/seaborn – visualization,- scipy – statistical tests.Python enabled efficient extraction of features such as HR, HRV (RMSSD), SCL means, SCRpeaks, and temperature changes.Statistical Analysis ToolsDescriptive statistics and inferential tests (e.g., paired/independent t-tests, effect sizes) wereperformed using:- SPSS, or- Jamovi, or- Python's SciPy.The choice of tool depended on the complexity of the analysis. Python was used for physiolog-ical signal processing, whereas SPSS/Jamovi was used for pre/ post-questionnaires, workloadassessment, and between-group comparisons.Operating System and Runtime EnvironmentAll VR experiences were executed on a Windows 11 machine running the SteamVR runtime.Physiological recordings were performed on a separate laptop operating the Biofeedback 2000Xpert software to avoid computational interference with VR rendering.3.4. System Implementation3.4.1. AI-Guided Voice Meditation PipelineThe AI-guided meditation in the VR condition is implemented as a pipeline consisting of threemain components: (1) text generation using a large language model (GPT), (2) text-to-speech(TTS) synthesis, and (3) audio scheduling and playback in the Unity-based VR environment.Meditation Script Generation via OpenAI GPTMeditation guidance was generated online using the OpenAI GPT language model (gpt-3.5-turbo) accessed via the official OpenAI Chat Completions API [Ope25]. Rather than playinga fixed, pre-recorded script, the system queried GPT at runtime and received short, adaptiveguidance sentences that were then converted to

speech.At the beginning of the VR session (Scene 1), a voice interaction sequence was used to initialisethe system. The TTS component first asked the participant: “Before we start, how is yourstress level: high or low?”. The spoken answer was captured via a VoiceCommandListenerand forwarded to the OpenAIManager, which parsed the response and stored it as an internalStressLevel enum (Low, Medium, High, or Unknown). This stress level was later embedded
into the system prompt sent to GPT, so that all subsequent guidance could be tailored accordingly(e.g. slower pace and more reassurance for “High” stress).The main meditation then took place in Scene 2, controlled by a Scene2Controller coroutine.Here, the current stress level determined a binary mode (high vs. low stress), which in turncontrolled the overall session duration (4 minutes for both modes) and the time gap betweenprompts (shorter gaps for higher stress, longer gaps for lower stress). The meditation flow wasmodelled as a loop of simple breathing cues (“Breathe in slowly.”, “Breathe out gently.”), butinstead of playing these fixed strings, each cue was sent as a user message to GPT together witha system message such as:“You are a meditation guide. Replies must always be under 15 words, one short calm-ing sentence, varied each time. The user’s current stress is: [High/Medium/Low].”The OpenAIManager constructed the JSON payload with this system instruction and the currentcue text, sent it to the OpenAI API, and parsed the returned chat response. From each reply, thefirst choice’s message.content was extracted as a short, GPT-generated guidance sentence.This sentence was then passed to the TTS component, which spoke it aloud and signalledcompletion back to the coroutine via the onResponseComplete callback. The coroutine waiteduntil TTS playback finished, then observed the configured silence interval (depending on stresslevel), and finally requested the next GPT line, continuing this cycle until the 4-minute meditationwindow elapsed.In this way, the spoken meditation guidance was not a single static script but a sequence ofdynamically generated, stress-sensitive micro-prompts produced by GPT in real time, while stillfollowing a deterministic temporal structure (intro, repeated inhale/exhale cues, outro) encodedin the Unity scene logic.Text-to-Speech (TTS) SynthesisSpoken guidance was produced using the Google TTS service. The selected meditation scriptis converted to speech with:- a neutral, calm voice profile,- a slightly reduced speaking rate to support relaxation,Audio Scheduling and Playback in UnityWithin Unity, an AudioManager component manages the sequencing of voice prompts. Themeditation session is modeled as a simple state machine (e.g., Intro → Breathing → MainMeditation → Closing). When the participant’s stress level has been selected, the correspondingsequence of audio clips is loaded, and each segment is scheduled with predefined intervals.The AudioSource component attached to the VR camera rig is used to render the TTS audio inspatialized form. This makes the guidance feel as though it originates from within the virtualenvironment rather than from an external device, supporting immersion and presence.
Stress Input(High / Low)GPT ScriptSelectionGoogle TTS(Audio Gen-eration)Unity Playback(Audio Manager)HTC Vive ProOutputFigure 3.3.: Vertical AI-TTS pipeline integrating OpenAI GPT script generation and GoogleTTS audio synthesis.3.5. Breathing-Responsive Light and Skybox AdaptationTo support paced breathing during meditation, the VR environment incorporates a dynamicambient-light system that is synchronised with the AI-generated voice guidance. Becauseparticipants meditate with their eyes gently closed, the skybox is configured with a high-intensitysolid colour, creating a soft diffused glow that remains perceptible through the eyelids. Thisglow serves as a non-visual pacing cue: the perceived brightness increases during inhalation andgradually decreases during exhalation, guiding the breathing rhythm without requiring visualfixation.The illumination behaviour is implemented through a custom Unity script (LightOnAudio),described in Section 3.4, which monitors the meditation AudioSource and interpolates the solidskybox colour intensity rather than a directional light. A tunable fadeSpeed parameter ensuressmooth, gradual transitions to avoid abrupt changes that could disrupt relaxation.Figure 3.4 illustrates the high-intensity breathing light used during the meditation phase.
(a) Dimmed skybox state ('breath out'). The sky-box decreases in intensity during exhalation,reinforcing slow relaxation breathing.(b) Bright skybox state ('breath in'). The solid-colour skybox increases in intensity, creating agentle glow perceived through closed eyelids.Figure 3.4.: Breathing-responsive skybox illumination. The VR environment alternatessmoothly between a brighter (inhale) and darker (exhale) skybox colour, synchro-nised with AI-guided meditation cues.The skybox behavior is governed by an

EnvironmentController script (see Section 3.4),which animates several shader parameters over the session:- emission color and brightness of the aurora bands,- global skybox exposure,- blend transitions between “inhale” and “exhale” color schemes.These parameters are interpolated using animation curves aligned with the meditation timeline(introduction, breathing guidance, closing phase). Initially, the skybox appears brighter withneutral tones; as the session progresses, colours shift toward cooler, softer gradients, reflectingthe intended transition from alertness to calm introspection.An example of the aurora-inspired skybox used during the meditation phase is shown in Figure 3.4a and Figure 3.4b.3.5.1. Multimodal SynchronisationBy coupling AI-guided audio, breathing-responsive illumination, and an emissive skybox, thesystem delivers a coherent multimodal cueing mechanism. Even when the eyes are closed,participants perceive a gentle pulsation of light aligned with the breathing guidance, reducingcognitive load and supporting stable attentional focus. The multimodal integration is designedto:- promote consistent breathing tempo,- reduce visual demand compared to object-focused VR meditation,- minimise mind-wandering through continuous but unobtrusive sensory input,- enhance subjective depth of the meditative state.Participant feedback (Section 4.4) indicates that most users found the through-eyelid glowcalming, whereas a minority reported brightness sensitivity, suggesting a need for adaptiveluminance control in future versions.2007-12-31/100/IN00/3333 23

3. MethodologyAI/TTS Med-itation Audio(AudioSource)LightOnAudio(Monitor play-back, fading)Directional LightIntensity (through-eyelid glow)Skybox Shader(Aurora emis-sion, exposure)Participant Perception(Breathing-pacedglow, calm ambience)Figure 3.5.: Visualization pipeline linking AI-guided meditation audio to dynamic ambient light-ing and skybox emission, resulting in a through-eyelid breathing cue perceived bythe participant.3.5.2. Integration with Physiological RecordingAt the start of each experimental phase (baseline, stress induction, meditation). During medita-tion session the Unity application and the Biofeedback 2000 Xpert system run in parallel. Later,these time stamps were aligned with exported physiological time series to support phase-specificanalyzes.3.6. Participants3.6.1. RecruitmentParticipants were recruited through university mailing lists. The recruitment targeted adultswithin the university community, without the requirement of prior experience in meditation orvirtual reality. Participation was voluntary and uncompensated.3.6.2. Sample CharacteristicsA total of 44 participants (7 female, 37 male) participated in the study, with an intended alloca-tion of approximately 22 participants per group (VR meditation vs. conventional meditation).Participants ranged in age from 22 to 35 years and included university students and staff. Basicdemographic information (age, gender, meditation experience, VR familiarity) was collectedthrough a pre-session questionnaire.2007-12-31/100/IN00/3333 24

3. Methodology3.6.3. Inclusion CriteriaParticipants were eligible if they met the following criteria:- Age 18 years or older- Normal or corrected-to-normal vision- Willingness to wear physiological sensors (Biofeedback 2000 Xpert)- No known medical contraindications for VR use3.6.4. Exclusion CriteriaParticipants were excluded based on any of the following conditions:- History of epilepsy or photosensitive seizures- Severe susceptibility to motion sickness- Cardiovascular abnormalities that interfere with physiological measurement (e.g., arrhyth-mia)- Skin allergies to adhesive electrodes used for SCL/BVP recording- Current treatment for severe anxiety, PTSD, or other conditions that may interact withmeditation tasks- Prior participation in a similar VR meditation pilot test3.6.5. Sample Size JustificationAn a priori power analysis was performed using G*Power 3.1 [FELB07] for an independent-samples t-test with medium effect size (Cohen’s d = 0.5), a two-tailed test, and a significancelevel of α = .05. The analysis indicated that achieving 80% power would require a total ofN = 128 participants (64 per group).This ideal sample size was not feasible given the time-intensive nature of VR sessions and therequirements for the physiological sensor setup. Similar VR-based psychophysiology studiestypically include 20-45 participants (typical for VR psychophysiology) [WWM20, XGH25,Zea19, Rea21].Accordingly, the present study targeted a realistic sample size of approximately 40-50 partici-pants and ultimately recruited N = 44 participants. A post-hoc power estimate for this samplesize produced an achieved statistical power of approximately 0.36 (see Figure 3.7), which isacceptable for an exploratory or pilot study designed to detect medium and larger effects. Fig-ure 3.6 illustrates how statistical power increases with total sample size for a medium effect(d = 0.5). At the actual sample of N = 44, the achievable power is approximately 0.38, whichis typical for exploratory or pilot studies of this scope.2007-12-31/100/IN00/3333 25

3. MethodologyFigure 3.6.: G*Power X-Y plot illustrating the required total sample size across different sta-tistical power levels for an independent-samples t-test with medium effect size(d = 0.5), α = 0.05, two-tailed. The targeted sample size of N = 44 correspondsto an achievable power of approximately 0.36.2007-12-31/100/IN00/3333 26

3. MethodologyFigure 3.7.: G*Power a priori power analysis for an independent-samples t-test with mediumeffect size (d = 0.5), α = 0.05, and two-tailed testing. With a total sample size ofN = 44 (22 per group), the achieved power is approximately 0.36.3.6.6. Ethical ConsiderationsThe study followed the university's ethical guidelines for human-subject research. Participantsreceived an information sheet explaining study procedures, risks, and confidentiality measures.Written informed consent was obtained before participation. Data were anonymized and storedsecurely in compliance with GDPR regulations. Participants were informed of their right towithdraw at any time without penalty.2007-12-31/100/IN00/3333 27

3. Methodology3.6.7. Experimental and Adaptive WorkflowThe experimental sequence consisted of four main phases: baseline relaxation, stress induction,meditation intervention, and post-assessment (see Fig. 3.8). At the beginning, participants wereseated comfortably and asked to relax for two minutes while baseline physiological signals wererecorded using the Biofeedback 2000 Xpert system. During this time, the procedure and studyobjectives were explained.In the second phase, mild stress was induced through a short gaming task using Flappy Bird,designed to elevate arousal and cognitive load. Heart rate variability (HRV) and skin conductance(GSR) were continuously monitored to capture physiological responses to stress.Following the stress induction, participants were randomly assigned to one of two meditationconditions:(a) Conventional Meditation: Participants listened to the same audio content used in theVR condition, but delivered without VR, in a seated position.(b) AI-Guided VR Meditation: An immersive environment built in Unity 3D. The AI voiceagent (powered by GPT-based dialogue model) greeted the participant and asked them toreport their perceived stress level ("high" or "low"). Based on this input, the AI generatedappropriate meditation guidance and adjusted the guidance and the lights will be lit up andfade away according to help participants in breathing(e.g., aurora brightness, backgroundtempo).After the meditation session, all participants completed the psychometric questionnaire batteryto evaluate stress perception, affect, mindfulness, workload, and user experience.2007-12-31/100/IN00/3333 28

3. MethodologyParticipantRelaxation Phase (2 min)Stress Induction (Flappy Bird)Random AssignmentConventional Meditation AI-Guided VR Meditation(GPT-based prompt selection)Physiological Recording(Biofeedback Sensors)Post-Session QuestionnairesFigure 3.8.: Experimental workflow showing baseline relaxation, stress induction, randomizedmeditation interventions, and post-assessment stages.3.7. ProcedureEach participant completed the experiment individually in a quiet laboratory setting. Thepractical steps followed the workflow shown in Figure 3.8: after consent and sensor attachment,participants first completed a short baseline phase, followed by the Flappy Bird stress-inductiontask and the assigned meditation condition (conventional vs. AI-guided VR). Immediatelyafterwards, they filled out the post-session questionnaires and were briefly debriefed.In total, each session lasted approximately 25-30 minutes.3.8. Instruments and Measures3.8.1. Psychometric Questionnaires- PSS-10 - Perceived Stress Scale [CKM83]- MAAS - Mindful Attention Awareness Scale [BR03]- PANAS - Positive and Negative Affect Schedule [WCT88]2007-12-31/100/IN00/3333 29

3. Methodology- SMS - State Mindfulness Scale [TB13]- IPQ - I-Group Presence Questionnaire [SFR01]- NASA-TLX - Perceived Workload [HS88]- UEQ-S - User Experience Questionnaire Short [SHT17]All scales were administered in English and scored according to their respective manuals. Relia-bility values from prior studies (Cronbach's α > 0.80) indicate strong internal consistency[CKM83,BR03, WCT88, TB13, SFR01, HS88, SHT17]..3.8.2. Physiological MeasuresIn addition to the questionnaires, psychophysiological signals were recorded and derived asdescribed in Section 3.3.2, providing objective markers of stress and relaxation across theexperimental phases.3.9. Data Collection and ProcessingAll sensor data responses were recorded using synchronized timestamps exported from Unity.All physiological signals were resampled to 1 Hz to enable alignment across modalities andreduce high-frequency noise.3.9.1. Preprocessing and Downsampling of Physiological SignalsPhysiological signals (BVP, SCL, TEMP, PVA) were exported from the Biofeedback 2000 Xpertsoftware as Microsoft Excel files in .xls format and subsequently converted to .xlsx usingLibreOffice to ensure compatibility with the

Python-based analysis pipeline.All preprocessing was performed in Python (Jupyter notebooks) using pandas and numpy. Foreach participant and session, the raw time series were loaded, merged with their correspond-ing timestamps, and restricted to the experimental phases of interest: stress induction, andmeditation.Because the Biofeedback system samples at relatively high frequencies (10-200 Hz dependingon the sensor), a one-second resolution was chosen for statistical analysis. The signals weretherefore downsampled to 1 Hz using non-overlapping one-second windows. Within eachwindow, summary statistics (mean or median, depending on signal robustness) were computed:- BVP-derived pulse waveform: mean amplitude per second,- SCL: mean skin-conductance level per second,- TEMP: mean peripheral temperature per second,- PVA: mean pulse volume amplitude per second.This aggregation preserves the overall temporal dynamics of stress and relaxation while reducinghigh-frequency noise and file size. The preprocessed signals were saved as clean .csv files (one2007-12-31/100/IN00/3333 30

3. Methodologyper participant) for subsequent comparison and HRV analysis.3.9.2. Validation of the Downsampling ProcedureTo verify that the downsampling procedure did not distort the physiological patterns of interest,raw and downsampled signals were compared in a separate Jupyter notebook. For representativeparticipants, segments from the baseline, stress, and meditation phases were plotted both at theoriginal sampling rate and at 1 Hz.Visual inspection confirmed that the downsampled curves closely followed the envelope of theraw signals, preserving phase-specific trends such as:- increased skin conductance and heart rate during the Flappy Bird stress induction,- reduced skin conductance and heart rate, and increased HRV, during the meditation phase.In addition, basic descriptive statistics (mean and standard deviation within each phase) werecompared between the raw and downsampled data. Differences were negligible (on the orderof rounding error), indicating that the downsampling procedure maintained the underlyingphysiological structure while reducing noise and data volume.3.9.3. Heart-Rate Variability (HRV) Feature ExtractionSeveral standard HRV indices were extracted from the BVP signal using the NeuroKit2 library.The analysis followed the guidelines of the Task Force of HRV Measurement (1996)[otESoCtNASoPE96].The following metrics were computed:- Mean Heart Rate (HR, bpm) - average beats per minute.- SDNN (Standard Deviation of NN Intervals, s) - reflects overall heart-rate variability.- RMSSD (Root Mean Square of Successive Differences, s) - an index of short-termparasympathetic (vagal) activity.- pNN50 (%) - the percentage of successive NN intervals that differ by more than 50 ms,also reflecting parasympathetic activation.- LF/HF Ratio - ratio of low-frequency to high-frequency spectral power, often interpretedas sympathovagal balance.These metrics provide complementary information about autonomic nervous system regulationduring meditation, particularly the balance between sympathetic and parasympathetic activity.The resulting metrics were assembled into participant-level summary tables and exported as.csv files. These HRV tables served as the input for the statistical analyses described in Section3.10, enabling direct comparison of autonomic recovery between the conventional and AI-guidedVR meditation conditions.2007-12-31/100/IN00/3333 31

3. Methodology3.9.4. Questionnaire Item Sets and Composite Score ConstructionAlthough the item pool was inspired by established psychometric instruments (e.g., PSS-10[CKM83], PANAS [WCT88], SMS [TB13], and IPQ [SFR01]), the present study did notadminister the full validated scales. Instead, short item subsets were selected to minimiseparticipant burden and to focus on constructs most relevant to meditation and stress recovery.Raw questionnaire data were exported from the survey platform with numeric Likert responses(1-5). Missing-value codes provided by the platform (-77, -99, -66) were treated as missingdata and excluded from score calculations.Composite indices were constructed as follows:- Perceived Stress (stress_score): mean of four items adapted from the Perceived StressScale, assessing feelings of unpredictability and lack of control prior to the session.- Negative Affect (affect_pre_neg): mean of four items adapted from the negative-affectportion of PANAS.- State Mindfulness (mindfulness_score): mean of items probing present-moment aware-ness and bodily attention during the meditation phase. These items were adapted fromstate mindfulness questionnaires.- Presence (presence_score): mean of three items evaluating immersion and the sense of"being inside" the virtual environment, adapted from the IPQ.- Satisfaction (satisfaction): single-item rating of overall satisfaction with the meditationmethod.For each composite, the mean of all valid items was computed, yielding a continuous scorebetween 1 and 5. These indices were used in all statistical comparisons and correlationsreported in Chapter 4.3.10. Data AnalysisAll analyses examined (a) physiological recovery from the

stress induction to the meditationphase and (b) differences between the conventional and VR-guided meditation conditions. Pro-cessing and statistical evaluation were carried out in Python using pandas, NumPy, NeuroKit2,and SciPy.3.10.1. PreprocessingPhysiological signals recorded by the Biofeedback 2000 Xpert system (SCL, HR, PVA, TEMP,MOTION) were exported as raw time series and downsampled to 1 Hz to ensure temporalconsistency and reduce noise. For each participant and each phase (stress induction, meditation),the first 240 seconds were retained to standardise window length. Mean values for each signalwere then computed for both phases.Heart-rate variability metrics (RMSSD, SDNN, LF/HF ratio) were extracted from BVP peakintervals using NeuroKit2 and summarised at the participant level [MPL+21].2007-12-31/100/IN00/3333 32

3. Methodology3.10.2. Change ScoresTo quantify recovery following stress induction, a change score (Δ) was computed for eachsignal:Δ = MeanMeditation − MeanStress.These values provide a normalised measure of autonomic adjustment, reducing inter-individualbaseline differences.3.10.3. Statistical TestingNormality of paired differences and Δ distributions was assessed using the Shapiro-Wilk test[SW65]. Depending on the outcome:- Within-group effects (stress vs. meditation) were tested using paired-samples t-tests[Fie13] or Wilcoxon signed-rank tests.- Between-group differences (Conventional vs. VR) were assessed using independent-samples t-tests or Mann-Whitney U [MW47] tests applied to the Δ values.Effect sizes were reported as Cohen's d [Coh88]for parametric tests and rank-biserial correlations[Ker14] for Mann-Whitney U comparisons. A significance threshold of $\alpha = .05$ was used forall analyses.3.11. SummaryThis methodological framework outlines the systematic process for evaluating psychologicaland physiological outcomes of an AI-guided VR meditation experience. The next chapterpresents the results obtained from these analyses and discusses their implications for adaptive,biofeedback-based mindfulness systems.2007-12-31/100/IN00/3333 33

4. ResultsThis chapter presents the physiological and behavioural outcomes of the study. Results areorganised into three parts: (a) heart-rate variability (HRV) during meditation, (b) within-groupchanges between the stress and meditation phases, and (c) between-group comparisons of thesechanges for Conventional versus VR meditation.Statistical tests were selected automatically based on distributional assumptions. Paired t-testswere used when normality was met; otherwise, the Wilcoxon signed-rank test was applied.Between-group comparisons employed independent t-tests or Mann-Whitney U tests. Non-significant results are reported as ns ($p > .05$).4.1. Heart-Rate Variability During MeditationHeart-rate variability (HRV) metrics were extracted from BVP signals during the 240 s medi-tation period. The key indices included mean heart rate (HR), SDNN (overall HRV), RMSSD(short-term parasympathetic activity), pNN50 (beat-to-beat vagally modulated variability), andLF/HF ratio (sympathovagal balance).Table 4.1 summarises the group means and standard deviations.Table 4.1.: Group means (SD) for HRV metrics during the meditation phase.Group HR (bpm) SDNN (s) RMSSD (s) pNN50 (%) LF/HFConventional 97.95 (23.00) 0.146 (0.052) 0.187 (0.074) 53.60 (15.81) 0.94 (0.61)VR High 111.90 (25.31) 0.119 (0.043) 0.141 (0.043) 52.11 (16.45) 0.85 (0.41)VR Low 87.23 (12.30) 0.153 (0.098) 0.179 (0.135) 48.13 (22.46) 1.53 (1.38)Overall, HRV during meditation did not differ significantly between groups ($p > .05$). SDNNand RMSSD values were numerically higher in the Conventional and VR-Low conditions,suggesting slightly stronger parasympathetic activation relative to VR-High; however, noneof these differences reached statistical significance. The LF/HF ratio exhibited considerablevariability, especially in the VR-Low group.34

4. ResultsFigure 4.1.: Group-level HRV metrics (mean ± SD) for the three meditation conditions.No statistically significant group differences emerged for any HRV metric. The HRV resultsare therefore interpreted as descriptive context rather than as primary evidence for differentialeffectiveness of the meditation modalities.4.2. Within-Group Changes: Stress vs. MeditationTo evaluate whether meditation induced measurable physiological relaxation, each group'sstress-phase values were compared with their meditation-phase averages. Table 4.2 summarisesthe direction and significance of these changes.Table 4.2.: Within-group comparisons between stress and meditation. Arrows denote directionof change (↑ increase; ↓ decrease). "ns" indicates non-significance ($p > .05$).Group SCL HR PVA TEMP MOTIONConventional ↑, $p < .001$ ns ns ↑, $p = .013$ nsVR High ns ($p = .065$) ns ↓, $p = .031$ ns ($p = .066$) nsVR Low ↓, $p = .004$ ns ns ↑, $p = .040$ ↓, $p < .001$Across all conditions, skin temperature increased during meditation, consistent with relaxation-related peripheral vasodilation and reduced sympathetic arousal. SCL responses differed be-tween modalities: the Conventional group showed a significant increase in SCL,

whereas the VR-Low group exhibited a significant decrease, indicating stronger electrodermal down-regulationin VR-Low.2007-12-31/100/IN00/3333 35

4. ResultsFigure 4.2.: Group-mean SCL over time (0-240 s) for all three meditation groups.VR-Low also showed a pronounced reduction in movement amplitude ($p < .001$), suggestingdeeper physical stillness. VR-High showed a significant decrease in PVA, which may be relatedto individual differences in breathing or sensor placement rather than a consistent group-levelrelaxation effect.To illustrate the thermal response, Figure 4.3 shows the group-mean skin temperature for Con-ventional versus VR (collapsed).Figure 4.3.: Group-mean skin temperature over time (0-240 s) for Conventional versus VR.All groups show a gradual increase in peripheral temperature, with VR tending towards slightlyhigher final values, although these differences did not reach statistical significance.2007-12-31/100/IN00/3333 36

4. Results4.3. Between-Group Differences in Physiological ChangeTo determine whether VR meditation produced stronger physiological regulation than conven-tional audio meditation, change scores (Δ = Meditation − Stress) were compared betweenConventional and VR (with VR-High and VR-Low collapsed). Table 4.3 summarises thesecomparisons.Table 4.3.: Between-group comparison of change scores (Δ). Only motion amplitude showed astatistically significant difference.Signal Conventional Δ VR Δ p-valueSCL 0.803 0.600 .487HR -0.096 -1.256 .708PVA 3.046 -4.153 .078TEMP 0.797 1.431 .473MOTION 0.026 -0.024 .005Motion amplitude was the only measure showing a significant between-group difference. Par-ticipants in VR conditions demonstrated a larger reduction in movement than those in theConventional condition, indicating deeper behavioral stillness and potentially greater attentionalimmersion.2007-12-31/100/IN00/3333 37

4. ResultsFigure 4.4.: Per-participant mean motion amplitude for Conventional vs. VR. VR resulted insignificantly lower motion during meditation ($p = .005$).All autonomic signals (SCL, HR, PVA, TEMP) showed no statistically significant differencesbetween modalities ($p > .05$), suggesting that both meditation types induced broadly similarlevels of physiological regulation within the time window of the experiment.4.4. Questionnaire ResultsA total of 46 valid participants completed the post-study questionnaire (VR meditation: $n = 23$;conventional meditation: $n = 21$). The questionnaire assessed baseline perceived stress,pre-pre-meditation negative affect, state mindfulness during meditation, perceived presence(VR only), and overall satisfaction with the meditation method.Table 4.4 summarises descriptive statistics for the main scales by mode of delivery (VR vs.conventional meditation).4.4.1. Baseline MeasuresBefore the meditation session, both groups showed comparable baseline perceived stress. AMann-Whitney U test indicated no significant difference between VR and conventional med-2007-12-31/100/IN00/3333 38

4. ResultsTable 4.4.: Descriptive statistics (mean ± SD) for the main questionnaire scales by mode ofdelivery.VR meditation ConventionalMeasure n M ± SD n M ± SDBaseline perceived stress 23 3.63 ± 0.91 21 3.55 ± 0.72Negative affect (pre) 23 2.94 ± 0.86 21 2.78 ± 0.76State mindfulness 23 3.42 ± 0.71 21 3.04 ± 0.91Overall satisfaction 23 4.04 ± 0.64 21 3.00 ± 1.34Presence (VR only) 21 3.34 ± 0.74 - -itation, $U = 246.0$, $p = .9238$. Likewise, negative affect prior to the session did not differsignificantly between groups, $U = 264.0$, $p = .6021$. These findings suggest that participantsentered the experiment with similar emotional states, so later differences are unlikely to bedriven by initial group imbalances.4.4.2. State Mindfulness During MeditationParticipants in the VR condition reported slightly higher state mindfulness during meditationthan those in the conventional condition (see Table 4.4). However, this difference did not reachstatistical significance, $U = 270.0$, $p = .5090$. In other words, there was a tendency for VRto support sustained attentional focus, but the effect was not strong enough to be consideredreliable with the present sample size.4.4.3. Perceived Presence in VRPerceived presence was assessed only in the VR condition. Overall, participants reporteda clearly positive sense of being “inside” the virtual environment (Table 4.4; $M = 3.34$,$SD = 0.74$ on a 1-5 scale). No equivalent measure was collected in the conventional group, sono direct statistical comparison is possible.4.4.4. Overall Satisfaction With the Meditation MethodIn contrast to the baseline measures, overall satisfaction differed clearly between groups. Par-ticipants in the VR condition reported substantially higher satisfaction ($M = 4.04$, $SD = 0.64$)than those in the conventional condition ($M = 3.00$, $SD = 1.34$). This effect was statisticallysignificant, $U = 354.5$, $p = .0054$, indicating that the VR experience was perceived as moreenjoyable and more worthwhile overall.2007-12-31/100/IN00/3333 39

4. ResultsFigure 4.5.: Sense of presence ratings for participants in the VR meditation condition.4.4.5. Differences Within VR: Low- vs. High-Stress SubgroupsBecause VR participants were split into low-stress and high-stress subgroups based on theirphysiological responses, exploratory analyses were conducted within the VR condition. De-scriptively, participants in the low-stress subgroup reported slightly higher state mindfulness andpresence than those in the high-stress subgroup, while overall satisfaction remained similarlyhigh in both groups.These subgroup differences are descriptive and were not subjected to formal hypothesis testing,but they suggest that lower physiological stress may support a slightly stronger feeling ofimmersion in VR.4.4.6. Correlations Between Questionnaire MeasuresFinally, correlations between the main questionnaire scales were examined (Figure A.28). Asexpected, baseline perceived stress and pre-meditation negative affect were strongly related,whereas state mindfulness showed a moderate positive association with overall satisfaction. Inother words, participants who felt more focused during the meditation tended to evaluate thesession more positively.2007-12-31/100/IN00/3333 40

4. ResultsFigure 4.6.: Overall satisfaction ratings for VR and conventional meditation conditions.4.5. Summary of FindingsThe combined analysis of physiological signals and questionnaire responses provides a clearerpicture of how VR meditation compares to conventional audio-based meditation.From the physiological perspective, all groups showed indicators of relaxation during the medi-tation phase. Temperature increased across conditions, and motion generally decreased, with themost pronounced stillness observed in the VR-low group. Heart-rate variability metrics (SDNN,RMSSD, pNN50, LF/HF) did not differ significantly across groups, suggesting that–within theshort time window of this study–VR and conventional meditation elicited comparable auto-nomic responses.Skin conductance level showed an increase in several groups, which is likely attributable to fingerelectrode dynamics rather than genuine sympathetic activation. Importantly, no measure indi-cated that VR meditation produced higher physiological arousal than the conventional method.Instead, the reduction in motion particularly supports the interpretation that VR encouragedstronger physical stillness and bodily engagement.The questionnaire results complement and extend these observations. Baseline stress and nega-tive affect were matched between VR and conventional participants, ensuring fair comparability.Although state mindfulness scores were slightly higher in VR, this difference was not statis-tically significant. In contrast, the sense of presence was markedly higher in VR, reflectingthe immersive nature of the virtual environment. Most notably, overall satisfaction was signifi-cantly higher in VR than in conventional meditation, indicating that users found the VR-basedexperience more enjoyable, engaging, and well-guided.Taken together, the findings show that both VR and conventional meditation produced similar2007-12-31/100/IN00/3333 41

4. Resultsphysiological relaxation outcomes, but the VR condition offered clear subjective advantages.Participants reported feeling more present, more engaged, and more satisfied with the experi-ence. VR appears to enhance the qualitative, experiential dimension of meditation–withoutcompromising physiological relaxation and, in some cases, even supporting more consistentstillness.This integrated interpretation suggests that VR meditation is a viable and appealing alternativeto conventional practices, particularly when the goal is to increase engagement, immersion, anduser satisfaction while maintaining comparable physiological benefits.Additional descriptive figures for all signals are provided in Appendix A.2007-12-31/100/IN00/3333 42

5. ConclusionThis thesis evaluated an AI-guided VR meditation system integrating immersive visual environ-ments, real-time voice interaction generated through GPT, and physiological monitoring via theBiofeedback 2000 Xpert system. The study compared VR meditation with conventional audiomeditation using both physiological signals and self-reported measures of mindfulness, affect,presence, and satisfaction. This chapter concludes by summarising the findings and explicitlyaddressing the research questions and hypotheses posed at the start of the thesis.Addressing the Research QuestionsRQ1: How does AI-guided voice interaction combined with adaptiveaudiovisual cues impact relaxation and mindfulness during VRmeditation?The AI-generated voice prompts and immersive VR environment were designed to supportattentional focus and relaxation. Physiologically, VR meditation did not produce significantlystronger autonomic relaxation than conventional meditation, as HRV, heart rate, skin conduc-tance, and temperature showed similar changes across conditions. However, VR participantsdemonstrated significantly reduced motion amplitude, suggesting improved physical stillnessand reduced

external distraction.Subjective responses further supported this outcome: VR users reported higher mindfulnessscores on average (though not statistically significant) and markedly higher satisfaction andpresence. These findings indicate that AI-guided verbal cues and immersive audiovisual feed-back enhanced the experience of relaxation and engagement, even when physiological differenceswere modest.Conclusion for RQ1: AI-guided VR meditation enhanced subjective relaxation, presence,and engagement, partially supporting the hypothesis. The improvement was visible mainly insubjective experience rather than in autonomic physiology.RQ2: What role do closed-eye visual effects play in enhancing perceivedmeditation depth?The VR system incorporated closed-eye visualisation effects (e.g. breathing light) designed tosupport deep focus during phases where users naturally closed their eyes. Questionnaire datashowed that VR participants reported significantly higher presence and immersion compared43

5. Conclusionto the conventional group. Open-ended responses frequently mentioned feeling “transported,”“calm,” or “more focused,” attributing these effects to the immersive visuals.While no physiological signal directly isolated the effect of closed-eye visuals, the behaviouralreduction in motion and higher subjective immersion suggest that the combination of visualsand context-sensitive AI guidance supported deeper engagement.Conclusion for RQ2: Closed-eye VR visual effects contributed meaningfully to perceivedimmersion and calmness, supporting the hypothesis that visual feedback enhances perceivedmeditation depth.RQ3: Can physiological and self-reported data effectively track meditationeffectiveness in an AI-guided VR environment?The study demonstrated that combining physiological and questionnaire data provides comple-mentary insights into meditation effectiveness. Physiological signals captured acute relaxationmarkers such as increased temperature, stable or reduced heart rate, and decreased movement.Questionnaire results revealed subjective states (presence, satisfaction, mindfulness) that werenot directly observable through biofeedback alone.The two data sources aligned in several ways. For example, motion reduction in the VR groupcorresponded with higher presence and satisfaction scores. However, autonomic markers such asHRV did not differentiate strongly between conditions, suggesting that short meditation windowsmay limit physiological detectability.Conclusion for RQ3: Yes, combined physiological and self-report data provided a valid andmulti-dimensional assessment of meditation effectiveness. This supports the hypothesis thatintegrating biofeedback with subjective reports is essential for evaluating VR-based meditationsystems.Overall InterpretationTaken together, the findings show that while physiological relaxation responses were similar be-tween VR and conventional meditation, the VR condition offered clear experiential advantages.Participants felt more immersed, more satisfied, and more positively engaged when guided byreal-time AI in a virtual environment. The behavioural stillness observed during VR medita-tion indicates that immersive systems may facilitate attentional stability even when short-termautonomic changes are subtle.Final RemarksThe results demonstrate that VR meditation, when combined with AI-guided interaction andmultimodal biofeedback, is a promising platform for delivering accessible and engaging stress-reduction interventions. Although the system did not show superior physiological outcomes2007-12-31/100/IN00/3333 44

5. Conclusionover traditional meditation, it consistently improved subjective experience and behaviouralstillness–two key components of effective meditative practice.Future systems may extend these findings through longer interventions, improved sensing tech-nologies, and more adaptive real-time feedback loops. As VR technologies continue to evolve,AI-guided immersive meditation holds strong potential as a next-generation tool for mentalwellbeing and personalised relaxation training.2007-12-31/100/IN00/3333 45

6. Future WorkThe present study demonstrated that AI-guided VR meditation is technically feasible and sub-jectively well received, but it also revealed several limitations and opportunities for furtherdevelopment. Building on the findings and on observations made during the user study, thissection outlines promising directions for future work.6.1. Adaptive Biofeedback with Modern SensorsIn the current system, physiological data from the Biofeedback 2000 Xpert were recorded andanalysed offline. GPT-generated guidance was therefore based on a pre-defined script and phasestructure rather than on truly live physiological changes. One natural extension of this work isto move towards a closed-loop meditation system in which modern sensors provide continuous,low-latency biofeedback directly to the AI model.Future versions of the system could integrate compact, wearable devices (e.g. wrist-based PPG,ECG patches, skin conductance sensors, or camera-based remote PPG) that stream heart rate,heart rate variability, respiration proxies, skin conductance level, or even

facial expressionsin real time. These data could be transformed into simple, interpretable features (such as“current arousal level” or “trend of relaxation”) and fed into the GPT-based meditation guidevia structured prompts or tool calls.This would enable the AI to dynamically adapt its behaviour, for example by:- delivering slower, grounding cues when arousal remains high,- prolonging calming phases if relaxation is still progressing,- offering reassurance if stress markers increase again,- or gently re-focusing attention if motion or micro-movements indicate restlessness.Such a system would turn the AI guide from a static narrator into a responsive meditation coachthat tailors language, pacing, and focus to each user’s moment-to-moment physiological state.Exploring the usability, robustness, and clinical relevance of such adaptive biofeedback loops isa compelling avenue for future research.6.2. Eye-Closure Detection and Comfort-Aware Visual DesignDuring the user study, several participants reported discomfort caused by the bright “breathinglight” used as a visual focus cue. Despite being instructed to close their eyes both by theexaminer and by the AI guide, some participants did not fully close their eyes or opened them46 6. Future Workintermittently, which led to irritation and distraction. This highlights an important practicallimitation of the current setup: the system assumes eye closure, but it does not actually monitoror respond to it.Future work could address this by incorporating eye-closure detection directly into the VRsystem. Many modern head-mounted displays already integrate eye-tracking or infrared-basedsensors that can estimate eyelid closure. These signals could be used to:- detect when the user keeps their eyes open during phases intended for closed-eye medita-tion,- automatically dim or switch off bright stimuli when the eyes are (re-)opened,- trigger adaptive guidance, for example a short reminder such as: “If it feels comfortable,you may gently close your eyes again.”Combining eye-closure detection with adaptive visual design would improve both comfort andimmersion. For instance, the system could gradually reduce brightness over time, or switch fromhigh-intensity focus cues to softer, peripheral visualisation once eye closure is detected reliably.Future studies could systematically compare variants of the breathing light (colour, intensity,frequency, and position in the field of view) and evaluate their impact on comfort, adherence toinstructions, and depth of meditation.6.3. Extended Protocols and PersonalisationAnother limitation of the present work is the relatively short, single-session protocol. Futurestudies should investigate longer-term interventions with multiple sessions over days or weeks,enabling the analysis of learning effects, habit formation, and more stable physiological changes.In such longer protocols, the AI guide could gradually adapt its language and difficulty levelbased on past performance, preferences, or self-reported goals (e.g. stress reduction, emotionalregulation, or sleep improvement).Additionally, combining the proposed real-time biofeedback and eye-closure monitoring withindividual preference profiles (such as preferred voice tone, music, or visual style) could lead tohighly personalised meditation journeys. Exploring how much personalisation is helpful, andwhere it might become overwhelming or counterproductive, is an important question for futurework.SummaryIn summary, future work should move towards more adaptive, sensor-driven, and user-aware VRmeditation systems. Modern wearable and integrated sensors can provide rich, real-time input toGPT-based guidance, enabling closed-loop adaptations to physiological state. At the same time,eye-closure detection and comfort-aware visual design can address practical issues observed inthis study, such as discomfort from bright stimuli and inconsistent adherence to instructions.Together, these developments offer a concrete roadmap for improving both the effectiveness andthe usability of AI-guided VR meditation environments.2007-12-31/100/IN00/3333 47

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